Notes: Krasnokutskaya (RES, 2011)

Identification and Estimation of Auction Models with Unobserved Heterogeneity

Contents

1 Big picture

- **Question.** Can we separately recover bidders' private information and auction-level unobserved heterogeneity in first-price procurement auctions, using only bid data?
- **Setting.** The paper studies first-price sealed-bid procurement auctions in which each bidder's cost has two parts: an auction-specific common component and a bidder-specific private component.
- **Core challenge.** Standard GPV-style first-price auction methods assume that, after conditioning on observables, all remaining bid variation comes from private information. That fails when auctions differ in ways observed by bidders but not by the econometrician.
- **Main methodological result.** The paper uses the *joint distribution of bids within the same auction*, together with a multiplicative factor structure and Kotlarski-style deconvolution, to nonparametrically identify the distribution of the common component and the bidder-specific components.
- **Key insight.** With unobserved auction heterogeneity, bids from the same auction share the same common cost factor. That within-auction dependence is exactly what makes identification possible.
- **Main empirical application.** The method is applied to Michigan Department of Transportation highway procurement auctions for bituminous resurfacing projects.
- **Main empirical result.** After conditioning on observable auction characteristics, private information explains only about **34.4%** of total cost variation. The rest is largely common auction-level heterogeneity.
- **Markup implication.** Average markups are much smaller once unobserved heterogeneity is accounted for: about **8.4%** under the preferred model, versus roughly **14%** under APV and **19%** under IPV.
- **Policy implication.** Reserve prices designed from misspecified IPV or APV models perform substantially worse. Depending on the benchmark, expected procurement costs can be about **9%–19%** higher than under the model that accounts for unobserved heterogeneity.
- **Economic takeaway.** In procurement data, correlated bids do not necessarily mean common values or collusion. They can arise because bidders share an auction-level cost shock. If we ignore that possibility, we overstate private information, overstate markups, and choose poor procurement policies.

2 Data and measurement (key objects)

2.1 Observed auction-level data

- **Data source.** Michigan Department of Transportation (DoT) highway procurement auctions.
- **Sample period.** February 1997 to December 2003.
- **Project type used in the main analysis.** Highway maintenance projects with **bituminous resurfacing** as the main task.

- **Raw sample size. 3,947** projects.
- **Observed variables.** Letting date, completion time, project location, tasks involved, identity of all bidders, submitted bids, and the engineer's estimate.
- **Institutional detail on reserve prices.** Federal rules say winning bids should be below **110%** of the engineer's estimate, but in the data some winning bids exceed that threshold. The paper therefore treats any binding reserve-price effect as small and works with a no-reserve-price model.

2.2 Markets, bidders, and cost objects

- **Auction format.** First-price sealed-bid procurement auction.
- **Private-values interpretation.** The paper chooses relatively simple highway maintenance projects because firms should know their own costs well, making the private-values framework plausible.
- **Source of unobserved heterogeneity.** Even simple projects differ in road curvature, elevation, traffic intensity, and preexisting surface quality. Bidders can inspect the site and learn these features, while the econometrician may not observe them.
- **Bidder asymmetry.** Firms differ mainly by size. The paper therefore allows two groups of bidders with different private cost distributions:
  - **Regular bidders:** large firms, defined as firms that consistently won at least $10 million in projects per year and had at least 100 employees.
  - **Fringe bidders:** smaller firms.
- **Core cost decomposition.** Bidder i's cost in auction j is

  $$c_{ij} = y_j x_{ij},$$

  where y_j is the auction-specific common component and x_{ij} is the bidder-specific private component.

2.3 Key descriptive facts

- **Number of bidders.** Between 1 and 11, with mean **3.4** and standard deviation **1.3**. More than **85%** of projects attract between two and six bidders.
- **Engineer's estimate.** About **75%** of projects have estimates between $100,000 and $1,000,000; about **5%** are below $100,000 and **20%** are above $1,000,000.
- **Winning bid relative to engineer's estimate.** Winning bids are typically below the engineer's estimate, and the gap grows with the number of bidders.
- **Money left on the table.** The difference between the lowest and second-lowest bid, normalized by the engineer's estimate, averages about **7%**. It declines as the number of bidders rises.
- **Regular bidders per auction.** Usually between 1 and 3; this rises only slightly with the total number of bidders.

2.4 Unobservables and timing

- **What bidders observe but the econometrician may not observe.** The auction-specific common cost component y_j.
- **What bidder i privately observes.** Its own idiosyncratic cost component x_{ij}.
- **What the econometrician observes.** Only submitted bids b_{ij}, observables z_j, and bidder identities/types.
- **Key information assumption.** The number of bidders is common knowledge to participants.
- **Across-auction independence.** The model assumes bids are independent across auctions. The paper later checks robustness to possible dynamic capacity considerations using a subsample based on backlog restrictions.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline cost structure

The model assumes a multiplicative cost decomposition:

$$c_i = YX_i.$$

In auction j, bidder i's realized cost is

$$c_{ij} = y_j x_{ij}.$$

There are two bidder groups. Let Group 1 contain m_1 bidders and Group 2 contain m_2 bidders. The model assumes:

- Y is the common auction-specific cost component;
- X_{1j} and X_{2j} are group-specific private cost components;
- Y and the X's are mutually independent;
- the densities of individual cost components are strictly positive and smooth enough for the inversion steps;
- the normalization is

$$\mathbb{E}[X_{1j}] = 1.$$

Interpretation. Multiplicative structure is convenient because it implies that the common component scales all bidders' costs in the same auction proportionally.

3.2 Expected profits and equilibrium bidding

If bidder i submits bid b_i and has realized private component x_i in an auction with common component y, then interim expected profit is

$$\mathbb{E}[\pi_i^y \mid X_i = x_i, Y = y] = (b_i - x_i y) \cdot \Pr(b_i \leq b_j,\ \forall j \neq i \mid X_i = x_i, Y = y).$$

A Bayesian Nash equilibrium is a collection of bidding strategies

$$\beta_i^y : [\underline{x}, \bar{x}] \to [0, \infty)$$

that maximize this expected profit for every bidder type.

Interpretation. This is the standard first-price procurement trade-off: bid higher and earn more conditional on winning, but lower the probability of winning.

3.3 A key scaling result

A central result in the paper is that if $\alpha_i(\cdot)$ is the equilibrium bidding strategy when the common component is normalized to $y = 1$, then the equilibrium bid function for any y is just

$$\beta_i^y(x_i) = y\alpha_i(x_i).$$

Interpretation. This means each observed bid can be written as a product of an auction-specific common factor and a bidder-specific "individual bid component." This multiplicative separability is what makes the deconvolution approach work.

3.4 Inverse bid functions and the first-order condition

Let $\xi_k(a)$ denote the inverse individual bid function for bidder group k when $y = 1$. Then the first-order condition for equilibrium bidding is

$$\frac{1}{a - \xi_{k(i)}(a)} = (m_{k(i)} - 1)\frac{f_{X_{k(i)}}(\xi_{k(i)}(a))\,\xi'_{k(i)}(a)}{1 - F_{X_{k(i)}}(\xi_{k(i)}(a))} + m_{-k(i)}\frac{f_{X_{-k(i)}}(\xi_{-k(i)}(a))\,\xi'_{-k(i)}(a)}{1 - F_{X_{-k(i)}}(\xi_{-k(i)}(a))}.$$

Interpretation. The left-hand side is the inverse markup term. The right-hand side is the hazard-rate-style object governing how fast the probability of winning changes as the bid changes, taking into account both same-group and other-group rivals.

3.5 Identification by using joint bid distributions

The paper does *not* identify costs from marginal bid distributions. Instead, it uses the fact that bids in the same auction share the same common component.

Observed bids satisfy

$$b_{ij} = y_j a_{ij},$$

where a_{ij} is the hypothetical bid bidder i would have submitted if the common component had been $y = 1$.

Taking logs gives

$$\log B_{i_1} = \log Y + \log A_{i_1}, \qquad \log B_{i_2} = \log Y + \log A_{i_2}.$$

This is exactly the structure needed for Kotlarski-style identification: two observables share a common latent additive component in logs. From the joint characteristic function of two bids in the same auction, one can recover the characteristic functions of $\log Y$ and of the individual bid components $\log A_k$.

The paper writes these as

$$\phi_{\log Y}(t) = \exp\left(\int_0^t \frac{\phi_1(0,u)}{\phi(0,u)}\, du - it\, \mathbb{E}[\log A_1]\right),$$

$$\phi_{\log A_1}(t) = \frac{\phi(t,0)}{\phi_{\log Y}(t)}, \qquad \phi_{\log A_2}(t) = \frac{\phi(0,t)}{\phi_{\log Y}(t)}.$$

Interpretation. Once the common log component is stripped out, the remaining variation reveals the latent group-specific bid components. Then the monotonic bid functions map those recovered bid components back into cost components.

3.6 Main identification theorem

Theorem 1 in the paper states that if the independence and regularity assumptions hold, then:

- the density of the common component $f_Y(\cdot)$ is identified from the joint distribution of any two bids from the same auction;
- the density of group-k private costs $f_{X_k}(\cdot)$ is identified if at least one of those two bids comes from group k.

Interpretation. The common component is identified from within-auction comovement; the group-specific individual components are identified from how that comovement differs across bidder types.

3.7 Why the standard GPV inversion fails here

Under standard first-price IPV methods, one inverts first-order conditions using the conditional bid distribution and density. But with unobserved heterogeneity, the bidder's optimal bid depends on the latent auction-specific Y. Since the econometrician does not observe Y, the needed conditional distribution of bids given $Y = y_j$ is unavailable.

Interpretation. The paper's main methodological contribution is therefore to switch from *marginal bid variation across auctions* to *joint bid dependence within auctions*.

3.8 Testable implications and distinction from APV

The model implies several restrictions:

- bid ratios and transformed bid components should satisfy conditional-independence relations if the individual cost components are independent;
- the recovered characteristic functions from different bid pairs should agree with one another;
- the inverse bid functions implied by the recovered objects must be strictly increasing.

The paper also shows that the model with unobserved heterogeneity is distinct from an **affiliated private values (APV)** model. Under unobserved heterogeneity, bids are conditionally independent given the common factor. Under APV, bids are affiliated. These are not the same set of distributions.

Interpretation. Correlated bids are not enough to conclude APV. The source of dependence matters.

3.9 Estimation algorithm

Conditional on bidder configuration and observed covariates, the estimation proceeds in steps:

1. log-transform bids;
2. estimate the joint characteristic function of bid pairs and its derivative;
3. recover the characteristic functions of the common and individual bid components;
4. invert via Fourier methods to recover densities of Y and A_k;
5. recover inverse bid functions $\xi_k(a)$ using the first-order condition;
6. map individual bid components into cost distributions F_{X_k};
7. renormalize to satisfy $\mathbb{E}[X_1] = 1$.

The paper proves that, under an ordinary-smooth tail condition on the characteristic functions, the estimators are **uniformly consistent**.

3.10 Monte Carlo design

The paper simulates auctions with costs

$$c_i = yx_i$$

using distributions similar to those estimated in the data. It sets the number of bidders to match the empirical configuration:

$$k_1 = 2, \qquad k_2 = 2,$$

and considers samples with

$$n = 250,\ 200,\ 150$$

auctions.

Interpretation. The simulation asks whether the deconvolution and inversion procedure is usable in finite samples that are not especially large. The answer is yes, except that precision deteriorates noticeably in the smallest sample.

3.11 Variance decomposition, markups, and reserve-price counterfactuals

Using the recovered distributions, the paper approximates cost variance as

$$\mathrm{Var}(c) \approx (\mathbb{E}Y)^2\mathrm{Var}(X) + (\mathbb{E}X)^2\mathrm{Var}(Y).$$

For homogenized costs, the individual component explains about **31%** of the variance. For total costs, once observable scaling factors are added back in, private information explains about **34.4%**.

Recovered average markups are:

- **Regular bidders:** about **8.4%**
- **Fringe bidders:** about **6.1%**

with regular-bidder normalized markups ranging roughly from **0.1%** to **25%**.

The paper also simulates inefficient allocation due to bidder asymmetry and finds:

- probability of an inefficient award: about **5%**;
- implied procurement-cost increase: about **2%**.

Finally, it compares reserve prices derived from four environments:

- unobserved heterogeneity with realized common shock known to the government;
- unobserved heterogeneity with the common shock unknown, so the government optimizes on average;
- IPV;
- APV.

Interpretation. The practically feasible reserve price under unobserved heterogeneity performs close to the benchmark, while IPV and APV reserve prices perform much worse.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the standard approach

- Standard first-price auction estimators assume all residual bid variation after observables comes from private information.
- If a common auction-specific cost shock is present, that assumption fails.
- Then GPV-style inversion of bids into costs confounds private information with unobserved heterogeneity.

4.2 What identifies the new approach

- **Multiplicative separability:** $B_i = YA_i$.
- **Within-auction dependence:** bids in the same auction share the same Y.
- **Independence assumptions:** the bidder-specific components are independent across bidders and independent of Y.
- **Monotone bidding:** recovered bid components map back to costs through increasing inverse bid functions.
- **Observed bidder types:** regular versus fringe status helps separately recover the two private-cost distributions.

4.3 Why the empirical application is credible

- The paper chooses simple maintenance projects to make the private-values environment plausible.

- It restricts attention to a narrow project type (bituminous resurfacing) so that technological heterogeneity across projects is limited.
- It uses homogenization to partial out observable project characteristics such as engineer's estimate, duration, district, and task counts.
- The remaining dependence in bids is then interpreted as evidence of an unobserved auction-level component.

4.4 Specification tests and robustness

- On a test sample of bituminous resurfacing projects with four regular bidders, the paper rejects IPV against unobserved heterogeneity with a **p-value of 0.03**.
- The corresponding test of unobserved heterogeneity against APV yields a **p-value of 0.52**, so unobserved heterogeneity cannot be rejected.
- Additional post-estimation tests do not reject the model's key independence assumptions:
  - conditional independence test: **0.52**
  - $X_i \perp Y$: **0.81** (bootstrap), **0.75** (subsampling)
  - $X_i \perp X_j$: **0.43** (bootstrap), **0.63** (subsampling)
- A collusion check based on Porter–Zona-style logic suggests the data are not obviously driven by collusive "phony bids."
- A backlog-based robustness exercise aimed at reducing dynamic capacity concerns yields similar conclusions.

4.5 Limitations

- The model assumes the number of bidders is known to participants.
- The common component enters multiplicatively.
- Auctions are treated as independent.
- Reserve prices are assumed not to bind in an economically meaningful way.
- The application conditions on a narrow set of projects and bidder configurations to keep the auction environment clean.

4.6 Relation to the literature

- Relative to GPV and related IPV/APV work, the paper adds unobserved auction heterogeneity.
- Relative to LPV, the common component here is *not* part of the bidder's private information in the same way; it is auction-specific and shared across bidders.
- Relative to instrumental-variable approaches to unobserved heterogeneity, the paper provides a direct nonparametric deconvolution strategy based on within-auction bid dependence.

5 Tables and figures

5.1 Figures 1–3: Monte Carlo performance

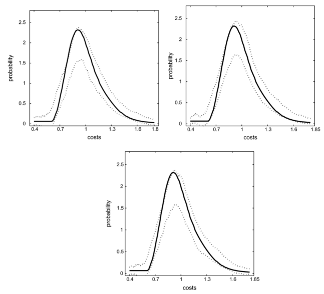

FIGURE 1

This figure demonstrates the performance of the estimation procedure in the simulation study. Solid lines depict the density of the unobserved auction heterogeneity component used to generate data. The dotted lines show 5% and 95% pointwise quantiles of the estimated distributions. The figures correspond to simulated data sets that include, respectively, (clockwise) 250, 200, and 150 auctions with two bidders from each group per auction

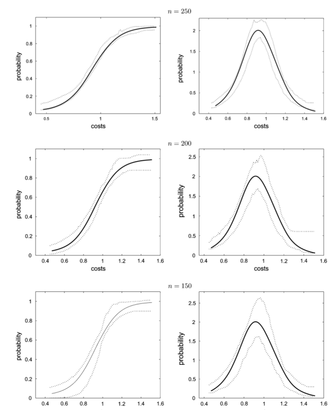

FIGURE 2

This figure demonstrates the performance of the estimation procedure in the simulation study. The solid lines depict the cumulative distribution functions (on the left) and the probability density functions (on the right) for the individual cost component of the regular bidders used to generate data. The dotted lines show 5% and 95% pointwise quantiles of the estimated distributions. The figures correspond to simulated data sets that include, respectively, 250, 200, and 150 auctions with two bidders from each group per auction

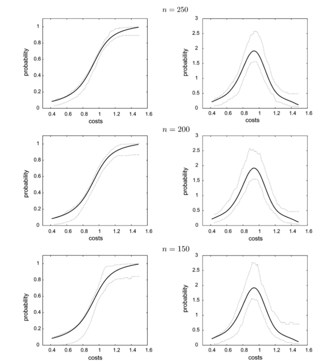

FIGURE 3

This figure demonstrates the performance of the estimation procedure in the simulation study. The solid lines depict the cumulative distribution functions (on the left) and the probability density functions (on the right) for the individual cost component of fringe bidders used to generate data. The dotted lines show 5% and 95% pointwise quantiles of the estimated distributions. The figures correspond to simulated data sets that include, respectively, 250, 200, and 150 auctions with two bidders from each group per auction

Read:

- Figure 1 studies the recovery of the common cost component, while Figures 2 and 3 study the regular and fringe individual cost components.
- The main message is that the estimators work well in moderate samples.
- At $n = 250$ and $n = 200$, the estimated cumulative distributions and densities stay close to the truth.
- Precision falls when $n = 150$, especially in the tails, but the procedure still captures the main shape of the distributions.
- This is important because the paper's empirical application does not rely on extremely large samples once it conditions on auction type and bidder composition.

5.2 Table 1 and Table 2: Descriptives and homogenization

Read:

- Table 1 shows the basic procurement environment: average number of bidders is modest, projects are mid-sized, and the gap between the lowest and second-lowest bid is economically meaningful.
- "Money left on the table" of around **7%** of the engineer's estimate signals nontrivial cost uncertainty.
- Table 2 reports the regression used to homogenize bids by observable auction characteristics before applying the main estimation procedure.
- The explanatory power of observables is meaningful but far from complete (R^2 around **33%** in the testing sample and **17%** in the estimation sample), leaving plenty of residual variation for the latent-component analysis.

TABLE 1
Descriptive statistics

Number of bidders	Overall	1	2	3	4	5	6
Number of observations	3947	71	673	1126	1026	365	192
Engineer's estimate ('00000)							
Mean	1·175	12·80	10·27	12·60	13·90	12·90	16·40
Standard deviation	4·660	2·35	1·41	3·02	2·26	1·79	3·39
Winning bid (hdrds. th.)							
Mean	1·175	11·10	10·00	11·80	12·90	11·80	15·20
Standard deviation	4·660	2·32	1·50	2·89	2·25	1·66	3·35
Money left on the table							
Mean	1·175	0·07	0·11	0·08	0·07	0·05	0·04
Standard deviation	4·660	0·05	0·08	0·06	0·06	0·05	0·04
Number of regular bidders							
Mean	1·175	1·92	1·43	1·65	2·07	2·16	2·29
Standard deviation	1·175	1·06	0·62	0·72	0·98	1·21	1·32

TABLE 2
Regression results

Variables	Test subsample	Estimation subsample
Constant	0·375	0·327
	(0·010)	(0·012)
Engineer's estimate	0·8413	0·8113
	(0·028)	(0·025)
Duration	0·0011	0·0015
	(0·0013)	(0·0011)
Tasks	0·0014	0·0012
	(0·0007)	(0·0009)
$(N_{\text{regular}}, N_{\text{fringe}})$	(4,0)	(2,2)
Number of projects	370	226
R^2	33%	17%

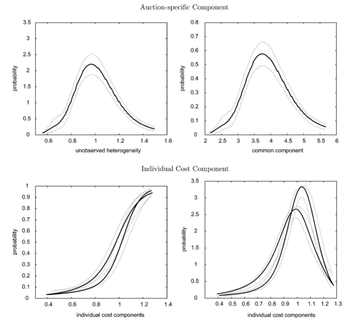

FIGURE 4
This figure depicts the estimated densities of the unobserved auction heterogeneity component and the common cost component as well as the estimated cumulative distribution functions (on the left) and the probability density functions (on the right) of the individual cost components. The dotted lines show pointwise 95% confidence intervals estimated through a bootstrap procedure

5.3 Figure 4: Estimated latent distributions

Read:

- The recovered unobserved auction heterogeneity component is centered close to one, with mean about **0.98** and standard deviation about **0.204**.
- After reintroducing the observable scaling component, the common cost component has mean about **$392,000** and standard deviation about **$78,890**.
- The regular and fringe private-cost distributions are similar, but not identical: the fringe distribution has a slightly higher mean (about **1.06**) and slightly lower variance.
- This figure is the clearest visual evidence that the model is not just mechanically fitting correlation in bids; it is decomposing that correlation into a common auction shock and group-specific private components.

5.4 Figures 5 and 6: Comparison to IPV and APV

Read:

- Figure 5 compares recovered bid functions. IPV and APV imply substantially lower cost levels and therefore much higher markups than the preferred model.
- Figure 6 compares recovered cost densities. Under IPV and APV, the cost distribution is much flatter and more dispersed.
- Quantitatively, the variance of the cost distribution is about **18%** lower under the preferred model than under APV and about **22%** lower than under IPV.
- These figures illustrate the central empirical point of the paper: ignoring unobserved heterogeneity makes costs look too dispersed and makes firms look too strategic.

5.5 Table 3: Specification checks

Read:

- Table 3 reports the p-values for the model's implied independence restrictions.

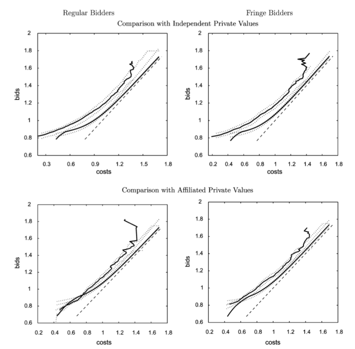

FIGURE 5

This figure compares expected bid functions estimated from the model with unobserved heterogeneity and alternative models (with APV and IPV). In each case, the lowest line shown is the diagonal. The dark solid lines correspond to the expected bid function estimated from the model with unobserved heterogeneity and the expected bid function estimated under the alternative model. In each case, a lower line corresponds to the bid function estimated from the model with unobserved heterogeneity. The figure also shows pointwise 95% confidence intervals estimated through a bootstrap procedure

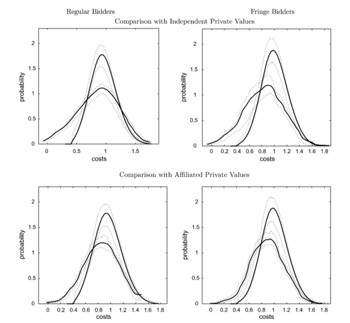

FIGURE 6

This figure compares the expected density of the total costs estimated from the model with unobserved heterogeneity and alternative models (with APV and IPV). The solid lines correspond to the expected densities. In each case, a tighter distribution corresponds to the density estimated from the model with unobserved heterogeneity. The figure also shows pointwise 95% confidence intervals estimated through a bootstrap procedure

TABLE 3

Results of testing procedures

	Test		p-Value
1.	$\frac{B_{i_1}}{B_{i_2}} \perp \frac{B_{i_3}}{B_{i_4}}$	Conditional independence	0·52
2.	$X_i \perp Y$	Bootstrap	0·81
		Subsampling	0·75
3.	$X_i \perp X_j$	Bootstrap	0·43
		Subsampling	0·63

TABLE 4

Reserve price

Medium projects: engineer's estimate = 4·0('00,000)			Unobserved Heterogeneity	Unobserved Heterogeneity, Expected	ipv	apv
1.	Probability of submitting a bid (expected)	$c_0 = 5$	0·29 (0·26, 0·31)	0·28 (0·25, 0·31)	0·07 (0·05, 0·09)	0·03 (0·02, 0·06)
2.	Expected cost of procurement (as % of c_0)		85·2 (83, 87)	86·5 (85, 88)	93·5 (91, 95)	96·8 (95, 98)
3.	Expected cost of procurement (as % of unh)			101·4	109·7	113·6
1.	Probability of submitting a bid (expected)	$c_0 = 7$	0·45 (0·43, 0·46)	0·42 (0·40, 0·43)	0·20 (0·18, 0·21)	0·04 (0·03, 0·05)
2.	Expected cost of procurement (as % of c_0)		72·1 (71·2, 73·5)	73·2 (72·3, 75·1)	83·9 (81·2, 84·7)	95·4 (94·3, 96·2)
3.	Expected cost of procurement (as % of unh)			101·6	116·4	132·4
1.	Probability of submitting a bid (expected)	$c_0 = 10$	0·58 (0·56, 0·59)	0·55 (0·54, 0·56)	0·20 (0·18, 0·21)	0·037 (0·03, 0·04)
2.	Expected cost of procurement (as % of c_0)		58·1 (56·2, 59·5)	59·1 (57·8, 60·1)	78·7 (76·5, 79·7)	94·7 (94·0, 95·6)
3.	Expected cost of procurement (as % of unh)			101·6	135·5	163·0

- The main takeaway is that the preferred unobserved-heterogeneity model survives these tests comfortably.
- So the within-auction dependence in bids can be interpreted as a common auction shock rather than as obvious evidence against the model.

5.6 Table 4: Reserve-price counterfactuals

Read:

- This table is the paper's most policy-relevant result.
- For example, when the outside procurement cost is $c_0 = 10$, the benchmark expected cost under the preferred model is **58.1%** of c_0. The practical reserve-price rule under unobserved heterogeneity gives **59.1%**, which is very close.
- By contrast, the IPV-based reserve price gives **78.7%** of c_0, and the APV-based reserve price gives **94.7%**.
- Expressed relative to the benchmark, the practical unobserved-heterogeneity rule is only about **1.6%** worse, whereas IPV is about **35.5%** worse and APV about **63%** worse in that case.
- The broader lesson is that misspecifying the source of bid dependence can seriously distort procurement design.

6 Short summary

- **Method.** The paper develops a nonparametric estimator for first-price procurement auctions with a multiplicative auction-level common cost component and bidder-specific private cost components.
- **Identification.** It identifies the common and private distributions from the joint distribution of bids within the same auction, using Kotlarski deconvolution and monotone bidding.
- **Application.** In Michigan highway procurement auctions, private information explains only about **34.4%** of total cost variation.

- **Main result.** Standard IPV and APV approaches overstate markups and distort reserve-price design because they mistake auction-level heterogeneity for private informational dispersion.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that first-price auction models with unobserved auction heterogeneity are nonparametrically identifiable under economically meaningful conditions.
- It also provides a constructive estimator that is uniformly consistent and feasible in moderate samples.
- Empirically, it shows that a large part of bid variation in procurement auctions reflects shared auction-level cost shocks, not just bidder-specific private information.

7.2 Economic intuition

- When all bidders face the same hidden cost shifter in an auction, their bids move together.
- Looking only at individual bid distributions makes that common movement look like extra private uncertainty.
- Looking instead at *joint* bid distributions within an auction allows the econometrician to separate common and idiosyncratic components.
- Once that separation is made, bidders appear less informationally dispersed and less highly marked up than naive models suggest.

7.3 Takeaway

This paper is important both methodologically and substantively. Methodologically, it shows how within-auction bid dependence can be turned from a nuisance into a source of identification. Substantively, it shows that procurement policy can go badly wrong if the econometrician confuses unobserved auction heterogeneity with private informational dispersion. In the Michigan data, accounting for the common auction component materially lowers estimated markups and leads to substantially better reserve-price choices.